

ActInf GuestStream #009.1 ~ "Collective intelligence and active inference"

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hello everyone welcome to the active inference lab today it is august 20th 2021 and we are here in guest stream number 9.1 with some awesome guests so we'll just start with introductions and then head into a short presentation before a question and answer period so if you're watching live definitely write questions in the chat so i'm daniel i'm a researcher in california and blue i'm blue knight i am an independent research consultant in new mexico and i'll pass it to raphael hi everyone uh thank you guys for having us over i'm rafael i'm also an independent researcher i have a day job at google that has some but not a lot to do with this topic and i've been working with pranav and jacob on this topic for about a year and a half now uh it's been a wild ride so right now go ahead hello everyone uh thank you daniel and pluto for inviting us i'm a phd student uh at carnegie mellon tepper school of business uh my areas of research are collective intelligence where does it come from building a causal model social technical architecture looking at collectives as adaptive systems um you know i was very interested in the uh active interest framework we met rafael i and jacob met you know across different ci conferences and got this project going so we're very excited to talk more about and learn from all the discussions that we have back over to you guys sounds cool so let's go to raphael's presentation and then we'll just have a bunch of time for questions so go for it all right before i get started can you all confirm that you can see me alright uh yeah we see the slides the slides perfect all right cool all right so um what are we doing here um so as we mentioned uh the the interest here is to try to understand better uh the phenomenon of collective intelligence and so two examples here what we're talking about and they're pretty different examples one of them is a pretty classic uh example it's the murmuration of starlings top left i know the bird that's solving a puzzle on the top right is a crow not a starling so i'm cheating a little bit but that's fine uh so uh the on the bottom the example is the discovery of the exposure which is an amazing example of kind of a large scale long-term scientific collaboration so those are both legit uh examples of collective intelligence and and yet the way

that we approach analyzing these uh things uh tends to be a little bit different so uh if you're doing something around complex aerodynamical systems or computational biology uh you tend to look primarily at the collective behavior and to be able to do that uh in a computable way you assume that individual agents are following simple rules to react to the environment and to their peers and if you're doing something like uh like anthropology or social science uh you're probably thinking of collectives that look more like the bottom and so you're focused on these really sophisticated agents that have these um complex social cognitive capabilities that are the thing that endows them with the ability to access collectives and you try to build a picture of the collective behavior kind of bottom up from the some of those interactions and both of our valid perspectives of course but they're speaking pretty different languages uh and we think that there must be a missing link here because if you think about us humans uh i mean i i definitely feel like i am at the same time an autonomous an autonomous individual and a member of not just one but a bunch of collectives right so i fluidly go in and out of uh companies or sports team or uh or uh uh polity and of course i'm part of my family so all these collectives can demonstrate intelligence behavior that that cannot be explained exclusively by aggregation of of the individual intelligence and yet uh the collective itself is a complex system that is composed of components that are in and of themselves highly autonomous uh complex adaptive systems as well and that's how we frame the the goal of our of our research we're looking at what you call the missing link which is a plausible theoretical description of the functional relationship between these two scales and uh in particular we think that the the missing link will be found by understanding what kinds of social cognitive features enable the formation of an effective collective within within these kinds of autonomous agents all right to do that uh we've been uh using active inference as a as a framework uh since uh the group here is probably uh way familiar with active inference already i'm not gonna go uh a lot into it uh just pointing out here that we're approaching this uh you might see the language that's a little bit uh different from from some of the canonical stuff that uh first and team put out we're using some different nomenclature but it's really the same principle we're all looking at an agent as a markov blanket and uh trying to to model exactly what's the generative distribution that that creates behavior in that in that agent and look at what happens when you put it against some true system dynamics that are that are happening in the environment and of course you you you run the the system as an agent model uh accepted in our case it's a multi-scale model right because we're interested both in what happens at the scale of individual agents and the global system for us it's actually quite uh interesting to try to decouple things and look at

what's going on in a very specific way modeling many many-to-many relationships uh is really hard so we really focused on the pairwise subsystems so our our uh model is really composed of a collection of pairs of agents they only have interact with each other within a single pair and we basically compose them the the what we call the system scale model bottom up from copies of these pairs of agents let's look first at the individual scale where we have um each pair is operating on the same physical environment they're sharing uh you can think of them as sharing one of these uh circles um if you're familiar with uh with the mcgregor model that has been published in 2015 it's uh it's a very simple very simple intentionally stylized uh active inference model that was originally designed to be analytically solvable actually so it allows us to have a really clear uh view of what's going on on the system as a dynamical system and uh we just added some minimal capabilities in that model so each of the agents that are marked out as little can think of them as little parts within this uh within the circle environment they each of them can only do a couple of things which is their capabilities they can only observe their their own position uh they can't directly observe the environment or that that is uh uh meaning they can't observe their own position they can only get a clue uh which is an on-off sensor generated by a chemical signal on each cell i'm getting a pin here from a friendly viewer also known as my wife saying that my image may be blurred can no it's good for others it might look a little different but i have a high resolution one saved locally no matter what the final processed youtube one looks like thanks youtube all right thanks youtube uh all right cool so as uh as i was saying the the main way that agents uh get information about their physical environment is by this uh sensory state which is a binary uh one on off sensor that's uh you can think of it as being generated by a chemical signal or a kind of smell on each cell and it's emanating from this from the source position source position um so that's pretty much it about the physical environment and agents can go uh left and right or stay stand so on that environment uh now talking about the relationship between between agents and their peer on a pair uh their pure impair is a funny thing to say but the agent can observe their uh their peer or their partner's position relative position meaning i'm one or two or however many uh um hops to the left or to the right they can be actually in the same position as well uh and they can also observe their their motion right there they're you can observe whether the the agent the other agent the partner has gone to left right or sits on the last um last iteration right um let's talk a little bit about uh reward functions uh the agents have independent reward functions that are two beats uh with one common peak and one private peak that the reason why that support is going to become clear a little bit later cool so that's it that's pretty much it about the

individual agents uh as i promised a really simple uh really simple model and the global scale as i said uh is a number of agent pair you can think of like 80 copies of these two pairs of agents meaning 160 agents in total or or whatever and uh from that perspective of the the higher scale the the markov blanket is actually composed of the of the collections of all of these pairs right so they are each interaction interacting with a um with a global environment and uh the cool thing is that we're we're um the way that the sensory inputs are relayed to each agent is as desires so you can think of it as the top level as kind of bringing in some uh some nudges some target states that are that are pretty much the the sort of smells that the agent picks up um all right the other important thing is that the optimal state for the system corresponds to the agent's shared target only but so from the system's perspective it wants all the agents to be in that shared target but they the agents themselves they don't know it uh why it's important is that it means that there's no exogenous incentive to to achieve the global goal so it's not like just for nudging we're asking all all the agencies the same thing and lo and behold they're going to do it all right and let's talk a little bit about how free energy minimization happens here uh on the individual level on the level of the individual agent uh we just assume that there's some process some neurological process within each agent that uh that does gradient descent that's the traditional technique uh but at the global level what we're actually doing is that the behavior that we're simulating at the collective of all these individual agents the the aggregate of that behavior is uh what we want what we would like to see approximate gradient descent so that's the thing that we actually want to demonstrate here we want to look at the behavior of the collective um calculate the the free energy the collective free energy which is f_{σ} here in this uh in this situation at the bottom and uh see if it's actually minimized if it's minimized then we uh then we can say that the system this this group of very simple agents as a collective it's uh performing active inference at the at the global scale or not and uh shock review what we find is that the extent to which the collective is actually able to do that um the free energy minimization depends on the social capabilities that we uh endowed these simple agents with um so we focused on two um cognitive uh capabilities or or or features that are stylized versions of of things that we traditionally look at in uh in social science uh the first one is what we call theory of mind and what we define that as that is the ability of an agent to put itself in the shoes of another and we actually uh try to model that in a pretty literal way uh so if you think of shallow versus deep representations of of the other you can think of a shallower presentation as like a mechanistic oh if they're going this way it means it means this rule-based kind of kind of kind of way which you you could

think of being an internalization of a rule that i think somebody else is following but we actually did hear some something quite different right so uh it was based on the this philosophy of active inference that uh means that the agent has every agent has a self-actualization loop and what we said here is hey uh the self-actualization loop the internal part of it needs to be um implemented by some kind of quote-unquote neuro neurological circuits and what if this exactly the same type of circuit the same structure were being used to model the partner so so that you actually also had a partner actualization loop so this is quite literally putting yourself in the shoes of of your partner right so uh you have your own beliefs about your and about your desires and you also have the literally the same uh kind of belief about the partner's position and their desires and you can use that plus of course whatever sensory or whatever information you have from the environment to predict the the partners actions is exactly the same way that you're predicting your own actions the only difference of course is that uh you don't actually control your partner so you're predicting that that's the dotted line from a2 uh out or from a1 out in the in the right so you're predicting how you're creating counter factuals about what your partner is going to do and then of course you can observe observe whether that prediction matches the the outcome in the next round and uh use that each partner uses that too to calibrate their model of the of the of their partner um and uh the other thing is you want those two predictions to be consistent between the cells right you wanna you want the beliefs about yourself and the partner to have some consistency and that actually helps you uh do what we call a triangulation right so you know uh you know very maybe very little about your physical environment but you're seeing your partner do stuff and uh and moving around and you're using that information to inform your your belief about your own position so that's the the the power really the power of theory in mind for our model and we're going to see how useful that is now uh the other one is uh go alignment i now realize that i've been speaking for a long time so i'm gonna uh go a little bit faster and so the other feature that we talked about is this thing called go alignment and it's really uh about the ability for an agent to adapt its goals to to match another agent's goals and uh this is a this is a bottom-up incentive incentivization mechanism don't remember what i mentioned that the system actually has an optimum but has no direct way to tell the agents about it so the assumption here is that to the extent that you can't that you have overlaps between the two agents uh goals and it turns out that those that the overlapping goals are aligned with the system optimum that provides a bottom-up way for for collectively optimal behavior to to to happen and we didn't model dynamic the dynamics of how that go alignment happens we just

assume that it can be flipped on and off just again for for simplicity and that's uh that's the thing that we'd like to explore further cool um so we did a virtual experiment we ran simulations with all four possible combinations of with or without theory of mind with or without uh goal alignment and and basically what we and the baseline here which is without your mind without golem in corresponds to a scenario where these agents are pretty much blind to each other they're not interacting uh with them with each other at all we can see each other but they're not using that information uh and we're gonna see how those features actually drive the the behavior of individual agents and also the system itself cool so basically what we find is that uh i don't think i mentioned this before but we define in each pair there there's a strong a perceptually strong and a perceptually weak agent uh the weak agent is pretty much following almost uh random noise that they have very very poor smell receptors uh but they still they can still use the the partner's uh position motion to triangulate right so the effect here that we see agent-wise are mostly what benefit do these uh social capabilities uh endow the weak agent with so we see that model with both together actually is the one where the performance of those the weak agents consistently improves uh really well to the point where you know it gets almost as good at finding an optimum as the as the strong agent and uh why is that so the the basic intuition is that theory mind by itself uh in this model is not able to improve outcomes that that much because um remember each agent has two uh two goals which correspond to the the two peaks here in the in the middle drawing on the left side and uh because there's this this great ambiguity right if i'm going to to assert in a certain direction uh what goal the it's hard to interpret that in terms of going to a specific goal if you don't if there's two possible goals where you're going so that limits how much useful information the agent can get from the from the strong and uh goal alignment itself uh doesn't help by itself because now you have a single reward target target but you you're still not uh super well defining where you're going but putting the tape up the two capabilities together actually does result in a grass skin improvement so oh i'll let it let that sink in for a second uh theory of mine knowing put putting yourself in the shoes of another plus uh having a goal alignment the capacity to to to share a goal uh it actually creates benefits for the the agents that are compensates helps a lot to compensate for deficiencies in the in the in the agent's ability to navigate its physical environment and uh that also is true uh looking at the collective so remember we were talking about whether these uh models can minimize the the collective free energy f_{sigma} so we're interpreting this uh these runs of all these runs of optimization for the individual agents as part of one single uh bayesian inference step for the collective right and

what we see here is if we're doing this right the graph should look like uh like a gradient descent down the the free energy gradient and this only really happens consistently towards an asymptote of zero in the fourth model where you have both of these social capabilities as we mentioned you know get some improvement uh you you have some abilities from these agents that are coming from their individual capabilities but when you when you put uh the social cognitive together with the with the uh capabilities about the environment that's when you get the ability to really minimize the the system free energy all right so that's what uh what we uh found in our paper and we think it's a pretty interesting and compelling uh example of being able to link uh still in a very stylized way but uh but very clearly link uh social cognitive capabilities at the individual level that are just that are not just simple rules right it's not about simple rule following but it actually has to do with interpreting the other and using that information and linking that directly to um to active inference at the collective level uh and where we'd like to take that i think the the four most exciting uh things that we've been discussing the last couple of months are first of all uh can you actually look at experiments with with humans and because you know these capabilities these software these social capabilities are things that you can measure uh through your mind and the ability to align goals for things that you can test out with human subjects and uh can you use our model to to represent and simulate and contrast with what happens in groups of of humans and see if differences in their abilities to to have theory of mind and to align on goals matches up with with what we're defining here as free energy minimization that's one one idea another one is you know collectives one of the things that collectives can can have is behaviors around shared resources and these resources endowed them with capabilities in different ways so that creates we think that creates sort of an on-ramp to understanding the the idea of economic value that's we've been joking and jokingly calling it for tsunamics this is uh the further down you go the more speculative we get but we think that active inference can actually help us understand this notion of of where does value come from is it really just uh related to scarcity or is actually also related to to the capabilities that it's endowing with different agents how these two things relate to each other so i think that's a there's a there's a powerful idea there um this notion of uh gnomics uh i wrote a little bit about that i think it's uh it's been starting to pick up scene in some areas and uh you know ultimately the uh all of us are embedded in one big collective which is gaia from gaia theory uh it's the planetary scale uh collective intelligence that some sometimes works really well sometimes it doesn't work really well can we use these sorts of ideas of multi-skill um active inference modeling to to improve how we

define and model uh the the behavior of that guy assistant finally going from the pretty big to the pretty small individuals and individual humans brain uh we've been looking at jeff hawkins's uh a thousand brain theory from his recent book and there's a lot of interesting ideas that look a lot like active inference but uh as far as you know uh nobody said try to model what what he says is the fundamental unit of intelligence which are these cortical columns in the new neural cortex nobody tried to model those using active inference and as the name implies the the main thesis of this theory is that the intelligence in our brains comes from the collective behavior of all these all these cortical columns interacting by a certain algorithm and so what it would look what would it look like to try to implement this uh this year in the context of of the brain and what inter-protocol agent capabilities are required i'm going to pause there i've i've overstayed my welcome i'm really curious to hear what kind of questions or or thoughts we have in our homes yeah thanks awesome thanks for the fascinating present you're muted then thanks sorry um great presentation so maybe we could start with just uh pravnav and then blue just what is bringing you to being right here and what are your first thoughts either being involved on the work or on the collective behavior side more generally blue are you going start like author's profession thank you thank you uh no i think uh so this work in particular uh like i'll start with this work as a context to set up the the bigger idea is a part of the thing that we i'm interested in we're interested together is figuring out one understanding causality of where does uh you know collective intelligence emerge from like what are the interactions between humans or humans and machines those things uh while you can have causal theory from a management social science perspective it's very hard to know that it is optimal like is that the outcome uh like a statistical you know mechanics approach to looking at ideal gas behavior and saying that oh the at the aggregate level this is how the gas should behave but you can look at the micro interactions uh underneath so i think for from my perspective active inference is like you know this may not count correct but from a social scientist perspective it is an a theoretical way or physics way of looking at uh the markov blanket like looking at the statistical mechanics without worrying about how exactly does theory of mind come about so as raphael explained we had a partner actualization loop we think the analog in social science is theory of mind uh and and what we start to demonstrate is that even if there are individual uh active inference agents they will need some form of partner actualization and in humans you can observe that as the capability of theory of mind to be able to you know use each other as sources of information so if i cannot work in my environment because my physical skills like looking at you know smelling the uh different chemicals at

different levels is low but my social perceptiveness is high so i can look at my partner and infer something about my environment then i can still work in the environment right but that alone is not enough for us to work as a collective so the free energy of the collective will not go down just if everybody can understand each other's perspective or step into each other's shoes so then we need something more so the second thing that we introduce and you know test out is some form of aligning goals like there has to be a reason to find you know common peaks uh again the current model does not say how it comes about that is a social science type of question because the causality like do we trust each other do we have a good communication processes or not like there will be accuracies and efficiency issues in communication themselves all of this is not you know aif doesn't worry about that but it tells us that these two capabilities at the individual agent level are important to get a collective outcome like i think from our perspective that is really the uh core contribution that we are saying that there are some cognitive capabilities that are beyond an individual aif model the way mcgregor had set it up that are needed for it to show up as a collective all right there's much more to be done here but we are starting to draw these patterns so the exciting thing that uh raphael started uh you know talked more about was can we map those which is uh experimentally if we run uh human agents in a similar environment and you know and ask them to do a similar task and we have aif agents working together can we create a mapping of what is the you know theory of mind index like the score for this person for this person on this scale because that would help us preemptively say that all right if you have a collection of people with this distribution of theory of mind like their ability to uh understand each other's like social perceptiveness which is a standard scale uh in social sciences i think it's called rme reading the mind and the eye that's a standard scale so like if we can create a mapping from a real uh indicator of theory of mind onto these things we can start having some predictions that this team is likely to succeed more than this combination of people right so that starts to open up the area all right now we have some interesting uh uh understanding of how the whole system works and can we draw it to real human so like this is very much the beginnings of a grander uh research statement and there is like research direction that we are trying to pursue and using aif in uh tandem with you know cause and theories or uh management or social science theories to figure out more that how can we you know help teams better or design teachers does that like is that enough like i didn't go much beyond that but like very happy to talk more on other kinds of words over to you thanks that was great um yeah i really enjoyed this paper from um my background is like biological sciences and so like i've done cellular

and molecular neuroscience like all the way through my phd until i kind of took the jump off the diagonal um so for me it's i'm not as well versed in the social science aspect but i think a lot about collectives and i'm super drawn to collective intelligence and even um hierarchical systemic organization right like so how you know bodies are built right from like molecules to cells to tissues to organs etcetera um and even like to societies and ecosystems um like the gaia theory that you were talking about earlier so so this paper was awesome for me i do have like quite a bit of questions um especially at like the social science front like in the beginning of the paper you guys outlined like a whole lot of different um you know potential ways that we operate like as multi-system um multi-agent systems and so like a lot of them you have theory of mind in there um and also the goal alignment but you also had like shared norms and folk psychology and so what what made you choose just those two as opposed to like testing all the other ones out or did you try others or did you not figure out a way to like mathematically get it together or what was your logic there expedience right now do you want to take that yes i think so we had about six months of discussion of what would be the scope of the paper and that's where uh because it was like so many things like as you can see here i found this and we are all very excited about looking at it at the level of christian but before we could jump on to any of those things i think it was important to boil down as to what is the key you know pieces that are needed before it can go to specifics so the key piece was at this point aif does a very good job of explaining like an individual interacting with an environment and how it operates yes we can go to norms yes we can go to these things but these are details about how does you know norms emerge and at the end of the day again i could be misunderstanding some pieces of it because i i'm coming from social sciences to active influence so pardon my uh pardon me errors there and please correct me my understanding is that aif does not give you a sort of a human level explanation it is the same kind of collective outcome that we would see if of cells clumping together we would not call it theory of mind in cells because i don't think we have a feature that says oh cells have theory of mind of other cells like it could be like it is still an analogy so aif is not telling you that this is the mechanism it's saying that you need this as a functional piece to it so that the energy minimization happens and that is demonstrated to the map so the reason we did not go into a specific implementation was because i think that would be the next you know set next set of steps we first need to say that at the abstract level at the you know baseline level concept level we need to have certain additions to an aif framework when looking at collectives as against uh so one way of looking at collective that we you know do in the systems free energy system level free

energy is that we look at the entire system as a you know aif thing and plots free energy but that is very different from being able to say that oh how are these two members or these 15 members within this system interacting with each other what are the capabilities that these people have that is different from this whole system as an you know active inference system and that that is where i think if i am not like i don't mean to complicate this uh or sound vague i think we first wanted to establish all right what is the minimum mechanism that we need if we do that in one paper at the simplest level then let's start digging into how does the goal alignment or norm uh formation come about so that can then be a parameter that certain teams or certain set of individuals or subgroups of individuals are better at forming norms while others are not you would expect these people to minimize their free energy as a subgroup better than this other subgroup so that would be an exciting extension in which you can go but we didn't want to jump there because it would be doing too much at least that's where we were at raphael is that accurate or did i completely [Music] from my side where where i where i got to to the beginnings of what we were discussing is i had this question in my mind of you know it takes a ton of cells to create a great collective behavior it's usually it's usually a pretty big discrepancy between the the different scales at which you know different levels of hierarchy are are are are set up if you think about a cell and then uh an organelle sorry an organ or clearly not a biologist but uh even in um in these examples that we gave with with starlings or other examples with other animals other than humans it looks like it like the the the behaviors of ant colonies and so on it's a it's a collective behavior but it's really driven by the the the statistical mechanical uh implications of of these relatively simple uh rule follow group following behaviors whereas sword i don't claim to have demonstrated it so i'll continue calling it a hunch the hunch that that we had is because humans have this more sophisticated set of abilities you know implicat implies some uh some some ability to do things in a non-rule following way meaning that they're not uh they're not cause effect relationships at the at the uh individual level uh that allows collectives of humans to be formed in a more uh in uh with fewer humans and with more of more of a fluid aspect and meaning they can the collective can also have more flexible behaviors again don't claim to have proven it but that's where i i i feel like the um the two the two capabilities that we that we picked came from because theory of mine true what distinguishes true theory of mind from just having a heuristic about how your partner is going to behave is really this ability to to put put what you're seeing uh in the world through the same the same active inference loop as you're used for for yourself which will call putting yourself in the shoes of another and that's like

inherently not uh not a simple cause effect thing uh the other one with goal alignment even though we didn't demonstrate it quite quite as as deeply i think it follows the same principle of you know it being this kind of like mutual adjustment of expectations that eventually gets you to uh to uh to a shared goal and so you feel like that that has the kernel of an idea about what makes human connective special awesome one comment and then i'll ask a question from the chat you kind of pointed out that there's a lot of modeling at the single agent level and then sometimes when you have a massive number of agents interacting you can almost take these mean field approaches or statistical averaging and then the kind of complexity is in the middle where there's small groups and reconfiguring so that was really interesting and also just the challenge of different words used for the same phenomena across scales or across systems i understood it as sort of like you were saying this is the continent of theory of mind these are some of the key words or the questions and then that's just a first connection and now you could reconfigure what the collective could do or do other experiments so here's a question from the chat they wrote a feature of the model is that order emerges through the bottom-up self-organization rather than top-down priors like most active inference models on morphogenesis however the goals were pretty much hard-coded into the model what self-organization did is only to select the collectively optimal decision among a set of possible decisions so i'd like to know more how the model improves on existing literature and importantly what it says about intelligence which is understood as the capability to understand the world rather than just optimize outcome given all right again yeah talk a little bit to that and then if you have anything to add um i think this uh this really points to to another limitation presented by time when we decided to to to publish this for for entropy we had this notion of being able to investigate how goals emerged at the collective level as well and uh that turned out to just not be feasible in our time frame but i totally agree with the critique that ultimately what you do want to understand is is also how goals emerge we talk about that a little bit in the paper i think the the one one possible approach is this notion of exploring states and uh as a collective and seeing what happens to individual uh outcomes and if if there is an evolutionary nudge that would that would that would align incentives naturally you know from the from the perspective of the of the various combinations of all individual incentives as well as if you think about social incentives they have these ideas of keeping up with the jonas jones's or or however there's a bunch of ways to model that and we think that given uh given that kind of evolutionary nudge towards uh go alignment we could see that emerge as a totally bottom-up uh model as well even having said that i don't think that the model is uh completely

top-down uh given it's not giving like so such strong constraints uh and uh the way we try to to tease apart that that thing is that this notion of everybody has two different goals and uh but the system has a has a single single optimum state and but it has it's not like directly telling it's not directly giving incentives to the to the uh individuals to to do it right it's just like quote unquote by coincidence the the shared goal asc is the is also the optimum for the system of course you know as we then as we showed if without a way to align on that shared goal it's just as likely that each each uh individual will just uh pursue their own uh private goal we did play around with some models where the the shared goal was actually a lower peak entering the reward function than the private goals and uh of course that meant that that absent goal alignment the agents were more likely to pursue the the the private goals and the shared goal and so you have a little bit of that flavor of okay i think we can uh we can at least say something about that uh about that relationship that is not just the traditional bottom-up composing goals and of course the units are gonna are gonna strive for us as well you have i think i'll jump in and add a couple of oh mostly how we arrived at this visions i think so several uh uh like i think great question this is the kind of we started with a complex model and you know evolved it into what's the simplest instantiation for the goal of this paper so let me walk you through a couple of different things which we started out with but we're now planning ahead on so we first started with the idea that okay if you're looking for a minimal mechanism we need something through which and i'm talking in analogy in like you know two agents teams terms not in uh aif terms uh we need the two agents to not only understand each other's goals or distribution of desires but also compile them right so the the simplest way we initially started with that we gave each individual their own twin beat or three beat or you know how it however peaked a distribution of goals in their environment and said okay you can just take them and combine them but what we started encountering in that piece was that should we be thinking about having an additional value for approaching shared goals that is when the peaks are combined if my partner also has this similar peak it should be given higher value so there is like there is a synergy bonus or there is a collective goal bonus that should be given that minimizes my uh you know selfish desire to go after goals that is only mine like my private goals so like we've played around this sort of decision space but we found out that i think we'll have to create another parameter on giving them how much is their preference for personal goals versus preference for collective codes so that's an additional parameter that can be added to the model to each individual that they have uh i i forget the name that you're like selfish buyers or like team buyers something like we created something of that sort uh and that

would determine so you could also end up in a place where me as an agent is always going after my personal goals but rafael as a team member is always going after team goals or shared goals like that would be the distinction that we'll be able to draw out that was not part of the scope of this paper that's why we you know removed away that's a thing and simplified it as everybody has only two peaks and it is very clearly known that this is the shared peak and this is the unshared to simplify the model and the explanation uh what you can do is make a general version of it that you can have a way of compiling them you can add on a parameter which can say that i have a dream team preference like i have the social science equivalent would be my collectivism orientation uh individualistic versus collectivism collectivistic orientations right like if i am very collectivistic i am very likely to choose a goal that is good for the team if i'm individualistic i am likely to choose a goal that is good for me right i'm not optimizing for the team so like those kinds of things can be built in hopefully we'll be able to explore that further but we realize that that orientation by itself is not required to minimally explain the emergence of characters right so one interesting piece again not explored in the paper was over here the idea was that both agents should end up at the shared peak so they are trying to seek the common peak what if so that is how systems high system performance was uh defined but what if it is a what you would call a disjunctive class that is there are multiple goals in the environment the team as a whole should hit all the goals as against compile on a single goal what does that look like that is cooperation to achieve all the our model currently cannot handle that it does not know how to coordinate separate goals that are contributing to a team over here the assumption there is synchronizing or converging onto a position is what is good collective behavior that is not true like as you try to you know do more analogies on which direction you want to grow the model that would be an extension that one has to think about that how are we defining what you know better collective outcome looks like like that depends on the environment you're working in depends on how the goals are broken down so those are the places that we would like to go so you're absolutely right that it is essentially in right now an optimal decision-making situation because the simplest feature the goal was not to make the optimization the complexity of the paper the goal was to say that you need a heuristic to find the common ground that's it and here is the simplest one that we presented does that make sense i think it brings us to this question about what does make the higher order group an active inference agent versus really what does make something specifically an active inference agent at any scale is it just enough to have input of information and policy selection isn't that just control theory more broadly so how do we um figure out who is and

isn't the active inference agent at the level we designed it at explicitly which was the lower level formally defined and then where really your claim is kind of building the bridge to which is statements about collective phenomena and collective behavior so maybe either the authors are then blue yeah and uh i'll i don't have a ton to add to that i just wanted to say that you know but i'm not i don't have a strong background in complexity to theory or or even in dynamical systems but i was i was reflecting on on those sorts of things and it struck me that this assumption of gradient descent uh you know the assumption that the optimization is even possible is such a kind of an ingrained thing when you think about this kind of modeling that exploding that assumption a little bit and just and focusing on on exactly this question of is it even possible for a for uh for a system that looks like this to actually like in in practice uh minimize this free energy functional right is it and uh i think it it again my day job has to do with some some issues of organizational dynamics and so on and i think that has a lot actually to do with with uh system dysfunctions because basic assumptions of what needs to happen in order for things to fall into place is or sometimes you know not trivial at all and uh can you know the efficacy of a team of the of the collective at doing that kind of of minimization which we really understand as achieving some kind of shared interpretation of the of the world is is really what's at stake here i also want to add really quickly that uh we cheated a little bit our system is not an active the the global system is not an active interest model it's just an interest uh model uh it again came down to expedience so it's the the collective as a whole is not acting back into the into the uh environment so i think there are bigger questions there around how does that bigger loop evolve and how does that influence the capabilities that that agents have right if you're like really trying to just to get your system to to act optimally on the world and the levers you have are the kinds of of social cognitive capabilities that individual agents have would that select for more for more agents with more theory of mind with more go alignment with the mix so that's the kind of things that we would expect to see with the broader uh dynamical lens at the system scale thanks blue yeah so um from the like complexity uh perspective and kind of piggybacking off of like the bigger system and what is the bigger system so i don't know if you're familiar with uh maybe integrated information theory of consciousness but it really looks at like to what degree the whole is more than the sum of just its interconnected parts um and then um there's there's that like angle but then there's also a recent paper by david krakauer that talks about the information theory of individuality and they really talk in this paper about how in order for a unit to be a new individual um there has to be like so and this goes even back to like the origins of life theories

right like so you start off as like a collective of whatever molecules and then what makes you form a cell right and so what has to happen in the integrated information theory uh paper um or sorry the information theory of individuality paper is that there's this bi-directional information flow from both the bottom to the top and then from the top to the bottom which you also see like in human organizations when like you get a group of individuals that align on some kind of collective goal like so somehow a leader emerges right like so so if everybody's kind of like well i don't know what to do and like you know how people are they act like that so then somebody will take charge and direct the group even in an emergent unstructured way there becomes this bi-directional information flow that makes something a collective unit and and i just wonder um like i i just find myself questioning whether you have that if you've built a new collective unit or not so and do you think you could and can i jump in quickly so you bring up fantastic points i think the way i have simplified and understood active inferences that you have an agent it is considered to have an active uh working in an active infrastructure paradigm if it is interacting with its physical environment to reduce its surprise right like it is trying to act on it build a hypothesis and then act on it as simple as that but when we try to move to collectives what we are changing here so like when uh so for example if you have to say is this single human an active inference agent we ask does it interact with a physical environment you know to optimize an outcome then we ask is this organization or a team acting as an active inference as a system you can say that oh this organization is interacting with its physical environment with the resource environment and doing things the distinction that we're trying to build is how do you go from here to here which means when there are multiple individual aif agents in addition to having a physical environment you also have a social environment that is where you you have to be interactive right so like now you are so the idea of theory of mind or goal alignment all of this results in the like from my perspective in the social environment place which you know leads to a system as a whole behaving such that it looks like it is uh you know an active inference agent it appears to be an active internet agent because it is able to very dynamically work on its physical environment but coordinate or combine its information that is coming from decentralized sources internally to get you know the you know minimum free type of action so now building from like looking back from there the idea of so we don't do in our current paper i think as rafael said we are not modeling the entire system as an af agent what we are saying is this is simply a statistical look so if you look at all the different runs and compile them together it would seem as if in most cases the behavior of the system is such that it aligns with reducing free energy it is not actually actively

doing things this is just a compilation or aggregation of the different you know simulation runs that we've done because you're not trying to build an ai model at home because at the whole level if you're trying to build an af model the model would look different you would not have separate individuals in the world uh they will have to have you know act as uh some somewhere you know joint at the hip in which there is perfect uh or there's some form of information integration going on so the the idea that i think you brought up that in an unstructured environment there's a leader developing and they you know bring it down so this fascinating idea and i think the way we were thinking about this was if we are going down the direction of when goals are merged do we have a bonus for convergence right where does that bonus come from so i think one way of thinking about this would be that that is the leader's job the leader's job is to change the weightage that each member gives to you know pursuing a common goal so if left to my own devices just you know carrying the previous example i might be very strongly focused on my individual goal that is not shared with it but if we have a leader amongst ourselves the leader is changing my the way i weight these different uh goals in my perspective and if they're able to convince me that the common goal would be a better outcome for you so like in in my type of theorization i would say oh negotiation and goal alignment so that would be that we are talking and we are saying that all right let me understand what are you after are you looking for status are you looking for exciting job opportunities are you looking for uh you know more higher pay let me understand what your underlying desires and motivations are and let me try to get them closer to what the common goal is like by your pursuing the common goal you will get x and that motivates you and you know from from like a very modelling perspective it changes the weightage that you assign to common goals and that kind of individual i would not say that it doesn't have to be a single leader it could all be decentralized that people are helping each other understand why you know following the common goal is better for each one of them that is a discovery process in that sense it is you know very directly looped that once a collective starts forming or once consensus start forming it becomes easier and easier or more more profitable for other individuals to join into you know the group goal because there is very highly high chance of the success or some other resources that you know you get as a payout like those kinds of directions we can think about doing again none of this is done in this model it's a fairly simplistic model to talk about let's think about social environment how do you go how do you bootstrap from aif in a physical environment to what do the agents do such that it looks like at the high level it is aif it is not at this point but minimum cognitive architectures can i ask you

for power yeah yeah um so right like in the global system that's not an active inference agent but like each individual agent is an active inference agent and then they form this like dyad so is the dyad like this meso at this meso level is the dyad an active inference agent also or no i think uh raphael feel free to disagree with me i don't think it is an active inference agent at the diet level because there is no there is no changes that are happening in how i infer about my goal states or how i like the only thing that is happening is now i have a secondary source of information so in the dyad up until now i my only source of information was the physical environment if i was low skilled like i think when raphael was presenting like the orange language the low skill low skill person they had very little uh they had a very low ability to understand where they are in where they are or are in that environment and hence they were not very successful in achieving their most of the time but the high skilled the blue line which you know came down really quickly uh had a very accurate understanding of where they are on the map so they could quickly traverse to their goal the thing that changed was by having high social perceptiveness now i have a secondary source of information about the environment my primary skill is gone like i cannot infer from you know the chemicals but i can infer from my partner's behavior so in that sense there is additional information that is helping the individual agent be a better af right because now they have a source of location of information that they can trust and you know build accurately on that said that source of information is not changing their own weighting of their goals and that's why there's no top down like there is no uh as a collective we are not trying to influence each other we are only taking independent decisions now assisted by a social environment also and that that much is enough to see collective uh outcomes emerge but if you are looking for more complex scenarios in which you are supposed to make a decision of should we you know break down across multiple tasks or what how many of these people should do task a and do task b or understand who is good at what those kinds of things will require more sophisticated machinery so it is i would not say it is a proper af well that to that i would add my counter find that uh if you define a proper af model as just any markup blanket then yes the dyads are are um markup blankets composed of the where the internal states are the union of the two agents and uh yeah even the even the bigger collective is just a very boring uh agent right the collective is an af agent it has its uh its inputs and it has its active state which is the null state right but you know proper there mean is it useful to think of it of that of it in those terms i guess it it depends right we want to we wanted to learn some salient facts about the collective we didn't quite ask that many questions about the diet as a as its own unit but we could so in

the diet formation did you guys play with like i know that you did the strong and the weak and you had like the strong one had strong environmental preferences and you had some parameters that you played with like and and the weak agent didn't have any like was really stupid about the environment but like could lock on to the partner and infer based on the partner so did you ever play with parameters like what if you make one agent um you know weak in both and one agent strong in both or or what what happens like did you mess around with those at all is it like the scale of it like as you were tuning to get like what was just right for you yes i think we're not can go into a lot of detail on that uh we did learn one thing very interesting which is uh you could very easily replicate the phenomenon of the blind the blind leading the blind where uh and we turned out it turned out that if uh if the strong agents was also very socially perspective it was very easy for that agent to lock into the partners behavior and try to learn trying to try to infer more than was actually uh actually legitimate from from what the the partner was doing which was a sports almost brownie in motion and that actually you know it can throw a poor agent into a tailspin so uh it is it is when you start looking at these uh at these more like as i said less less rule following behaviors they are pretty sensitive to to to kind of this balance of parameters well that is exactly it i think a fantastic question blue uh and rafael has captured the essence of it is there is a combination so like we had the entire like i was doing robustness things and like there's an entire space what are the different combinations and how do they pan out the most striking example was when the strong agent has a very strong social perceptiveness also uh they even if they've achieved their target they start to move away from that because they're you know trying to get something they're getting confused about their own state because they think that the other member is still trying to move towards their goal state so either there must be some other goal state or like this is me putting explanation on top of the behavior that we saw so like we did not see a stabilization coming up now what this tells us it tells us a very important thing that our model is not complete at all right it simply says that you need these two features but these two features are not enough to see a fairly sophisticated version of collective behavior in which there is uh error correction at that level and and from my perspective again and i know that uh raphael and i uh you know have been going at understanding active inference and how we define it and that's why i come out on the side of the collective is not completely an active insurance model yet because it does not have error correction uh and how do we get there like there is more who wants to be built into that how is it different from if you have both all the agents in your team are very good in their physical environment so that's like having an all-star team we know sometimes

all all-star teams fail can we you know bring up that kind of that niche of behavior now all of these questions are unanswered and extremely exciting you know important areas to go into uh but all of that and that's what you know i come down and we started with norms and we thought that we should have this and we should have that but we boil down to okay what is the minimum thing that we need to say without this nothing works so the collective does not emerge even in minimal circumstances without these two uh in play if i could give one comment on that and it's such an interesting tension that your two answers brought out where if the system is designed as an active inference agent it's a no-brainer yes it is and then this debate which we've seen play out you know increasingly across so many different areas okay it's a system that wasn't explicitly designed as an active inference agent whether it directly computationally emerges from a designed agent or whether it's a performer on a stage or some other social system of governance and then there's the realism question is it actually quote an active inference agent versus the instrumentalism question can we use this framework statistically to do inference and just like the distribution doesn't have to be perfectly normal in nature for the use of a t-test there's ways to use active inference very usefully so it's so cool to see like a sandbox for exploring some of these fundamental questions about collective behavior which often times it feels like every single species or every system they have their own programming language or they have their own norms or the edges are transient so they go down the rabbit hole with the math that way instead of stable edges so what's the framework that's going to have the flexibility to actually cross these different bridges within the collective intelligence or collective behavior area so it's an awesome good question and i think that it goes to the heart of like local i don't know if you guys haven't read um the three body problem where they have uh you know uh spoiler alerts there's uh an alien alien species and you know uh they have this big computer that's made up of soldiers with flags uh you know the soldiers with flags make up a computer that's by definition it's a computer because it was designed you know they're they're following rules that that caused it to to compute uh certain functions right and you know that's that's how we design computers now there may be uh certain you know turing machines that emerge that are that are uh that we can even sometimes even prove that they are uh universal sewing machines and that's pretty awesome right and you know that's another way of saying that they are computers and i think there's the same uh same sort of uh question going on here with uh with active inference where you know is it a markup blanket yeah you can and draw the the the lines around it it's a mark of blanket is it's to what extent it's performing free energy minimization depends on how you define the generative density how

you define the the internal state so you can calculate the free energy functional and and yada yada and the proof is inputting right when you look at it does it like does it actually look like it's optimizing that that free energy function sometimes the answer to that is really hard to get or only feasible to get if you make some pretty uh aggressive assumptions as you're saying about uh mean field um and so on so it's it's at the heart there's uh there's an epistemological presence and almost an ontological one i'm going to ask a question from the chat this is from dean who wrote is there a difference between a self-actualization loop and what might be described as a sense of self-awareness and what is the pursuit of shared goals dependent on if common interpretation and translation such as symbolic manipulation is not generating a common context thank you that's an awesome question i don't pretend to have any answers to it i'll give you my uh sort of opinion i'd also love to hear from the others uh i think um my mental model of this is hey uh these agents are reasonable they only have to the extent that you can say that they have uh beliefs or opinions about anything they have beliefs or opinions about two things right their their uh their own position and their their partner's position in particular they don't have and these are literally just uh uh numbers from 1 to 16 right so there's two probability distributions in particular they don't have anything that we could associate with the self as a concept right because they're only looking at two positions right now i think there's definitely uh it's it's not trivial to say well what is uh does self-awareness mean a symbolic sense of self or is it just the epic phenomenon of having beliefs about some aspect of itself i'm not qualified to judge that but in my opinion what the agent has just information about a specific aspect of itself it feels like without the symbolic without like having a sort of more sophisticated causal model that has opinions about this opinions about that and opinions about how these other things can act and one of those things is the self which is representing is like a pointer to pointer to self i don't think that qualifies itself awareness awesome these are fun discussions blue daniel you needed oh it was on the stream but not agency so blue oh um i will uh follow up to that question actually so you have like this self-actualizing loop with the agent and then you have this like other act actualizing loop right and i kind of looked at it like your maybe your self-actualizing loop is like your iq and your other actualizing loop is like your eq maybe like or or either way um but but what i really thought about is like you know we are by nature like such incredibly selfish creatures and so like it seems that you like maybe gave equal weight in the model like can you maybe dig into like because in in reality we don't give equal weight like we would give the self loop way more priority than the other loop and was that a parameter that you played yes like i have a lot more

to say but like rough i'll go first and then that is yes but if i go ahead please yeah no no i'm just going to say yes i think the the primary question that we were looking at when we think about the these these loops if if you think about the just the basic function of of taking some input from the environment forming some uh some belief states that are just just really like a collection of of numbers and then uh and then acting on it uh that can represent pretty much anything that's what we were when we talk about a partner actualization loop you can really just think of it as two copies of the same circuit one of them is wired wired out into my own actions and the other one is not the one that's wired out of my own actions is the cell phone self-actualization group just by virtue of the fact that it's completing the the it's the thing that's completing the the markup blanket uh the the other one is is wired out into nothing so it's just generating predictions about my my my partner's behaviors but it's it has the same formal structure internally to the internally to the agent uh now i think the the part where it gets interesting is that uh consistency requirement where we uh uh we're talking about like we have uh we have beliefs about the about the the other and as as the predictions are that we're generating about their actions are related to those beliefs but i think what we're asking is really about uh what matters most to me right is it about uh am i paying more attention to my actions and my goals or to to my partner's actions and to and to their goals right so i think what the definition of matters is really about what's influencing my actions that is primarily about my beliefs but they are informed about my beliefs of it's primarily about my beliefs of my position but they're also informed by by my beliefs about my partner's position which is that process of triangulation that we talked about and that's the parameter the theory of my parameter that we were we're squeezing out so it is of course uh you know it's very simplified but in principle that's uh that's a range that can go anywhere i think we did play around even with uh with having some uh anti-social behavior uh programmed in uh and you know limited amount of interpretability yeah i think uh blue very insightful question again and uh like i think we were also thinking about as the self as being iq because it is individual skill and the partner actualization group being more social intelligence because it is about understanding the other person's interaction we sidestep that problem of how do you weight one over the other by choosing uh you know the space and again like i think this maps on very well with your previous question that oh what happens if both of them like if uh an agent is good at both right we see blind leading the blind which means that as soon as you see that diversity in agents that can be really good at both or really bad at both you'll need some form of weighting mechanism that to what extent do i value information that i have received from my primary skill versus i

value information that i've received from my social environment and by social environment now as you can imagine if you have more than one partner you'll get you will also get conflict information because the partners might be moving differently because they are differently able right so at that time we cannot sidestep the question of how do i weight the social information coming in from different partners so there has to be a trust or a credibility index that i have to maintain internally so this is very close to uh concepts like a transactive memory system in management literature right where we are trying to set up that oh i know that uh raphael is an expert at skill a and that's a credible source so even if i find a new team member who is equally skilled at uh you know uh the same skill okay i would more like i'm more likely to go to rafael for help when i need it because he has credibility he has some history so there is an index that i maintain internally a meta memory of mapping of who has what skills and how credible they are so like you cannot sidestep that question as we start building and working towards bigger collectives right so you are spot on with that and that will start to happen even at the simplistic level of this discrepancy can be created between my primary uh information source and my social information source we did not like as like we played around with oh let's give it antisocial like they always think that their physical thing is better this led us to having conversation like do they have meta memory which is like do they have self awareness that they are low skilled in the physical environment as compared to their partner right so this all started building more complex things like in my other stream of research i built those kinds of agent-based models which are social cognitive architectures that you give them a memory they give them a meta memory you give them meta attention you give them a reasoning a way to map who is working before how much do i trust them so like and that's where these ideas were flowing in from that all that complexity we cannot sidestep that complexity if we want to build you know that sophisticated behavior that was not the goal of the paper and that's why we didn't like we just removed all that like we just sidestepped that complexity by saying let's look at what is the minimum gains that can be had by doing x or y but the next step really if you are trying to build a more sophisticated behavior outcomes right not just minimizing system level free energy would be to poke in that direction so brilliant question like i'm like this is actively of how we've been thinking and you know making the model more complex and then simplifying this is the exact tension and interesting insights that you would get there so very happy to uh you know if you have more ideas on oh this would be something that would be nice to pursue i'm all yours you know one one thought that that brought to mind was when there's just a differential equation model like some kind of smooth line that's going to be

generated the model can't really fail like you could fail to read in your data set but it's not like the line's ever going to break so then people are wondering about how well does it fit and they kind of focus on the performance and then with agent-based modeling it's always understood that there's actually only a few islands of parameters and of course the choice of which parameters to include from which effective collective behavior is going to arise like it's not just enough to be an ant forager not every foraging strategy works and the one that works in the desert is not gonna work in the jungle so it's actually just these islands of stability and then you bring up all these questions like well what's the resilience of it to being perturbed those are the kinds of questions that we care about with the real world systems and it's almost like the affordance from a scientific and a modeling communication perspective of having a modeling approach like linear regression that pretty much accepts any kind of data set without really questioning it it leads to different questions that are being asked and it leads to different uses of the model so it's just so cool that those kinds of questions are arising and i'm sure they come up in interesting ways in your own research so we'll keep on yeah yeah this is super exciting because uh that's the direction that uh sorry did i speak over you raphael no no so this is clearly as it has been communicated this is super exciting to me because i think uh a big question that we've not asked yet is what is the goal of this kind of a modelling like there's one way to look at active inference can do $x y z$ like from my perspective the goal is okay how can we have these social cognitive models and active inference models go hand in hand so that we understand something about real teams and real teams have you know as you know you really elegantly put they have non-linear behaviors they have islands of places where they'll behave in a structured manner and suddenly it breaks because one parameter moved out of that island to be able to get there we need some minimal machinery so i think that is the space where i live in when we uh you know with anita and with chris riedel uh and i think i've not worked directly with david angle but like anita has pre-work with like that is the direction that we are pushing in in that space is like oh let's find this island of area where if meta tension and meta meta memory is you reasonably developed or there's diversity within the team how does that where are the stable coordination zones there right right so that is one way but in aif at this point we did not even have the minimal machinery that i think we could understand and interpret so we started like what's the first building block that we need to build and the confession is i don't think i completely understand aif rules i think they're fluid like how do you define an aif like raphael was bringing up if it is only a mark of market then yes it is if it is this then it is that so this could be my shortcoming because i'm not red enough or it could be

the field is actually really updating itself so so this is that's why this is exciting like let's build that but for me my eye is always on the like how do we pull it out into is this applicable like we don't want only equations that can never be wrong in a matter of time you don't get non-linear dynamics out of it directly sorry yeah awesome thank you yeah and then agree and i think this uh perhaps you you eluded that and i want to pull it uh pull it up it's it's really about the the scientific endeavor as a collective active inference uh behavior right if you think about how how scientific modeling in the in the kind of linear paradigm works it's about you know establishing uh universal truths that are not contingent uh you know that are you know contingent only on uh on a very very specific very explicit set of of assumptions that are knowable within that you know you pull it out uh you can explain it to an alien or you pull it out and write it into stone tablets and the next civilization can read it and and so on right and that's that's not what we're doing here right i think we're feeling our way as a as a collective towards a more useful more um more interesting and applicable way to to understand the real systems that are around us and then to maybe to learn something to help them function better so that's that's probably working at it um i'll ask a quick question from the chat because i think it applies and then blue so dean wrote where would the authors see citizen learning organizational structures blind leading the blind or basic research at scale no it's not a buzzfeed article from 2027 it might be maybe you're seeing a glimpse of the future i don't know enough to answer brandon you guys want to take a step but now if you're muted yeah then sorry yeah i was just responding to dean's comment it's a such an insightful comment there's like one hopes that this basically scale it is it is like it brings up the uh thought experiment that will democracy thrive if uh you don't have independent thinkers right like so if it is blind reading the blind that nobody is actually thinking everybody is just based on secondary information then it would look one way and it would be the same at the aif level it's the same behavior or it is independently thinking and basic research at scale and you get completely different outcomes in both cases so like i i don't think i have good answers there but i think very very legitimate question uh something that we can figure out i think that's that's model that's modelling and it relates to the discussion earlier about convergent collective behaviors versus divergent so whether it's like honeybees deciding on where to relocate or some other type of consensus protocol you want convergence but then you talked about an example where there was preference to be on separate niches and it's also important to have divergence like in discourse and in other areas and creativity so how to design that space is so critical i i think i think uh your spotlight but there is value in both of those see convergence is

important to get the outcome like the bees have to decide on where to go finally but divergence is important to assess the accuracy of the information so to the extent uh or rather to explore the space right accuracy and exploration of the space itself so to the extent it is blind leading the line i would say that is biased towards convergence which means the system is opening itself up to not settling on sub sub optimal outcomes uh because everybody agrees very quickly on what the solution should be uh lack of creativity that kind of thing if you are too divergent and you're not able to converge then the system is opening itself to not getting the rewards because nobody's agreeing on anything so you need both and that's what i think the independent research and the blind linear brand are fantastic things i don't think can resolve the issue unless there is one agent that is always understood to be they have the most accurate information always and the entire system understands that right like it could be like a small family universe's or whatever mom says is correct so all the children's are lined up because they are not trying to question they agree and trust that that is true and in that case convergence is great anyway um and actually to connect that to some of the analytical components of active inference in the free energy minimization there's one term that's often framed as pragmatic value and then the other one is the exploratory the epistemic component you know and accuracy minus complexity these different framings that we've seen and so it's so cool that you described the need to have a balance of the converging and the diverging collective algorithms not worrying about whether it is a formal active inference at the higher scale but just some of the same principles which is that you have to have some target seeking or signal detecting aspect and then some aspect that prevents at the very best premature convergence if not actually keeping options open to remain resilient so those are some of the the principles of collective systems that kind of go beyond even active inference alone and just it doesn't matter at that higher level exactly what it is and so really interesting stuff yeah so if you look in the literature uh in the management so you could look at mega and even corrective and uh complex adapter systems from holly arrow like they have a book on groups as complex adapter systems and like one of the key things that you know in in a separate line of research i build on that is there are two functions that any group has to do one is efficiency the other is maintenance the idea of efficiency is that the resource that you have you have to deploy very well and get the maximum out of it the idea of maintenance is that oh you need to choose the right goals so that the system can survive right because if you're very efficient but you're going after the low rewarding things then the system will not thrive on the other hand if you're choosing the right goals but your wasting away the resources and not applying them efficiently

the system will not survive so those two things together are functions of any group or any collective so i think again like for me it's like they're looking at the same phenomena from two different sides one is coming from what's the mechanics of it how do we achieve that so you look at all the social social cognitive architecture collective memory collective attention collective reasoning the other is like does this look like an aif and can we model it mathematically without worrying about what exactly is the mechanism right so it can prove as a ground truth to this team has way more to achieve than it is currently achieving because an ideal aif model can be better performing right and you're missing something so there is some losses happening in the processes in the groups right it could be we're not understanding we're not choosing the right strategy or we are biased towards convergence whatever that is right like that's something to be diagnosed about the team and that therein lies a practical application like you have to figure out from here to here what could be you know going wrong so yeah all right the thing that i want to say i wanted to add that is if you look at gaia theory uh and if you shake it hard enough and it looks like a dynamical system and you're throwing the the uh ergodicity assumption yeah i know maybe from the from the whole whole system throughout it's like billions of billions of years uh time scale perspective and you can back out and say yeah it's a single it's a single active inference agent and therefore there is one you can back back it into like one big system goal for all time that is only ever visible in retrospect from uh from the perspective of like some of the the time keepers in the multiverse or whatever right uh we hear simple agents uh try to do our part don't have access to that and we have to do with more proximate goals right these are often going to be at odds these are often going to turn out to be wrong and there's always this tension between how you construct uh the goals are you constructing them for for exploitation or for for exploration at what at what level how can we can in turn internalize internalize the work of other uh systems around you to to get towards more of a uh more bang for the bug for everybody and so on uh but it's all it's all really imperfect and i think that's what the the question trying to finally tie it back to that question on on citizen learning it's you know from uh from a long run perspective probably it's to some definition it works out okay but what okay means and uh you know in the long term we're all dead it's uh the devil is definitely in the details on that one nice that reminds me a lot of bucky fuller and humans role as sensemaker locally and making it work for everybody so it's really cool to see that kind of come back to active inference through design science and other areas so blue so i have like a small technical question and i'm just gonna load it up followed by my big question so pranav i am curious about your like social

science mapping and meta mapping and memory and meta memory just like how much compute power that takes like is it is it monstrous but then i'll i'll add another a couple couple more things so i have done a little bit of a agent-based modeling if you like need somebody to interrogate you over your model parameters i'm super good at that so i will harass you any time um and then uh the real question that i have is maybe because of the nature of agent-based modeling maybe it's you know you always have to define the parameters of the system and so we're looking to create a model where essentially the rules are broken right like when you make a collective the collective has broken the rules it's done what the individual agents it's not collective behavior it's something new right and so we don't know what that newness is going to be and so maybe agent-based modeling is is maybe the wrong tool like how do you allow in an agent-based model where you define all the parameters how do you allow for this like newness to emerge again blue fantastic question i'm so happy to keep listening to your questions they're just very insightful uh i i'm not sure if this is if i've understood the spirit of your question but i think based on what i understand i would say the use of agent based modeling is predicated on the goal that you have then in a social science or sort of like social theory perspective uh we are building a transactive systems model of collective intelligence that is we are saying that collective intelligence emerges out of uh the formation and co-regulation of processes that involve collective memory collective attention and corrective reasoning so intelligence is an outcome of these three functions but these are not independent as in every person has that that is true but there is coordinated so collective memory which is called transactive memory in the literature emerges out of the skill based interactions that coordination that people do same thing with attention same thing so if we're building that kind of a model the role of an agent-based approach to this kind of theorizing is essentially to say that okay i have this fantastic narrative theory that people understand are aware of each other's skills meta memory right and they use that information to make allocation choices that is if i have a certain task and it requires help with x skill who do i go to or who do i assign this to in the team so like if that is the theory how do we know that this is enough this is parsimony so this is the idea of generative sufficiency right like are these minimal processes enough to explain the emergence of your collective memory right like will a flock emerge if each bird is each void is given these three rules right so and that kind of research question or you know building of that model is important to basically give some credence to these are the minimum social cognitive processes that are important to give you this outcome so we are not trying to code flocking we are only giving individuals certain capacities and the flock is emerging so

that this is minimally enough now is that flock adaptive to uh let's say a predator approaching them maybe not because we've not told them how to react to a predator right so in that sense we now know there is a boundary condition that with only x rules it can you know handle high levels of workload but it cannot handle high levels of uncertainty like those kinds of things so computationally how difficult or how complex the thing depends on how many features you are trying to build into it the model that i've you know developed for my dissertation and i'm you know presenting uh as i'm going on the market uh is fundamentally saying that at the simplistic level give each individual three capacities of having skills and meta memory understanding you know my team members their skills having attention or focus like ability to work on something and understand how other people are working on different things so like what priority that is how important it is for them which leads to the meta reasoning which is i have selfish motivations and i know that other people also have selfish motivations but is the task aligned with you know each other motivation so i should send this task to somebody for whom this is already aligned so they'll get to it faster and that will benefit the team it's like none of this is a centralized uh here's the leader who knows everything and is making the allocations right this is not top down command control this is very decentralized how will an intelligence at the collective level emerge to the extent these members are able to figure these things out so that that's that's the building of the model that's you know headed towards that direction so i think what it becomes useful is under what circumstances the useful question to us is under what circumstances is meta reasoning or you know collective memory important so like this is the question that is very different from saying that or do we need this parameter or not and how computationally intensive it is the the answer lies in okay what's the environment what's the so like as you know daniel was pointing out there are islands of operation does this island in a phase space land in a place where these parameters are important or not so those are the kinds of questions you can answer with that and i think that would drive the choice of parameters i would not i would not try to say that or it should be a all encompassing model it should be the simplest model uh if anything i've tried to always say that we're trying to go to the simplest model that explains her part very clearly does that answer the question i i suspect it does not yeah so so um definitely like with flocking and like i've seen similar models of like fish schooling and and stuff like that um so that as a model of maybe collective behavior versus collective intelligence or like has something actually has something new emerged in the flock or in the school of fish and so my my point is is that when you just give these agents their individual parameters how do you know when there's another level like we're looking for

another level that does active inference so so not like a yes to collective intelligence you can even collective intelligence and collect behavior can be modeled with a group versus the individual agents but when do you have another level like in the hierarchy when do you when do your cells become a tissue when does your group of people become you know a squad or a team or something like this so that that's what i don't know if agent-based modeling is suited to to determine or not well that's a good question i think it depends on so it's partly a definitional issue what do you mean by another level right so intelligence the way it is defined at least in that literature collective intelligence is an ability to adapt to changing environments so as the task changes in its complexity that is uncertainty is changing or the level of knowledge interdependence or the level of workload is changing over time the team is able to seamlessly adapt its internal coordination processes to maintain performance so it is not just because the task has changed in its complexity so the environment is uh moving around just because the environment moved it should not be so specialized that it starts to fade it is able to adapt to it so if intelligence is defined that way to what extent can the you know the team regulate internally to maintain performance at the highest level possible then i would say yes it is emerging because the the diagram that you show right like essentially the plot that you show is a team that does not have these coordination mechanisms is failing like is dropping in performance anytime there is a perturbation while a team that has these mechanisms is like seeing a momentary dip but is able to maintain high levels of performance so it is it is a relative thing now if you change the composition the internal coordinations would be different because now it's a different set of people trying to coordinate but it would not invalidate the idea that a different set of people still have collective intelligence or not because it's given by given our internal resources how are we adapting to the changes like if you're able to track their so in that sense i think agent-based model does a fantastic job of showing you that we are able to see this emergent behavior of maintaining high level performance uh or not right and that's where you you go into the mechanisms that this mechanism is essential for this outcome versus not does that help it does it does but it's it's the question of at what scale is it an active inference agent right so like is the team the active inference agent or is it the each individuals is it like their collective active inference versus the team collective i can see daniel like oh wait i was just gonna say that yeah to speak to the question of novelty there's the question of generating novelty and then responding to novelty and in the case of generating novelty especially for humans we haven't really talked about stigma g niche modification cumulative culture so many avenues from collective behavior to more niche modifying algorithms that

maybe explaining some of the phenomena that we are interested in here but then this question about responding to novelty and like how do you deal with a threat that maybe hasn't been seen before or a failure mode that's never been observed or rare it kind of just reminded me of homeostasis which is generalizing across specific situations to help you keep target variables within a preferred range and then instead of just reacting to changes it makes one move towards anticipating which is the whole area of variously like cybernetics and signal processing and so those features of responding and anticipating and planning agent-based modeling is a great place to computationally and physically test that clearly active inference at the very least can be sketched out over a lot of these systems and then it will be a debate always like about the details and the mechanisms like you brought up but it's just to have a framework that we can integrate scales and just nest them cleanly is at the very least a path towards answering some of these questions if not philosophically at least empirically raphael thank you yeah and i think about the philosophical aspect also uh as backing down from unless you're looking at the gaia theory thing and you don't believe in such a thing as uh you know selection multiverse uh thing uh so if you back down to like subsystems of the big thing whatever the big thing then you can back down into functions right and you can use that as your benchmark for for for what performance or success or intelligence uh looks like and deconstruct uh down match that up with you get from from the bottoms up from the bottom of you and yeah i guess the the uh the interesting question is when and where does it actually do the the bottom up and the top down approach line up you know there's i think in general there's no answer there from uh from from an ecological perspective i think there's some some pretty clear patterns of functional specialization and then and uh some of them are hierarchical some of them are you know tend to be like almost hierarchical or hierarchical with a little little bit of splint sprinklings of of uh horizontal conditions and maybe that's a that's a useful you know philosophical approach to to to look at things that from from the instantiation at least from the instantiation of oh the kinds of systems that were interesting interested in it's when does a group of cells become become a tissue but it's also what does it what are the conditions that a thing needs to meet uh to be to be meeting the function of a tissue and to be uh worthy of being called the tissue in that in that functional context right and then you can talk about well what what if that tissue is not made of of cells but of like nano nanobots or whatever and so on yeah i do yeah i don't think it is a it is an either or type of question i think it is more of uh the two approaches of like coming at it from a mechanistic perspective versus coming at it from an active inference perspective are complementary they're trying to look at the same

phenomena but estimate them differently so the idea of uh i think then started mentioning that morphogenesis and morphostasis i think that that's like when novelty happens how do you respond to that novelty is part of you know one term in the aif equation and it is also a way of looking at saying that oh will the system adapt to it or not right so there's a mechanism of how it adapts and then there is the equation version of it if it adapts well it would be reducing its uh free energy right because that is what we are able to estimate i think both of them are very complimentary approaches what so one way i think i look at it is how do you know that the adaptation from a mechanics mechanistic shampoo that is correct or wrong right is it maladaptive is it going to lead for a better outcome or is it going to lead to a worse outcome one way of looking at it is is that reducing free energy or is that increasing free energy of the system right now that is where you can marry the two things and forecast compliments that or there are certain adaptations because as an agent is sitting inside the system you know not knowing the bigger picture there is no way you know uh whether these you know uh variations that you're trying are good or bad like you need a system level out so i think that is how i think they're complimentary again i don't have good answers yet because that's the area of research that you're trying to figure out but i the fact that that is where it could go we could combine the two approaches to get more insight on the both the mechanism and the overall outcome this has been great any last thoughts or any time you're always welcome to join again and it's just super interesting to hear about we hope that people watching it live and replay could think about some of these questions but anyone's welcome to give a last thought otherwise this was just a great first appearance with you both and we hope to see you again yeah just thanks this was super fun to read and to discuss and i think um i think that the theory underlying the model is hot and super on point and building it out and flushing it out into a more thoroughly um you know developed model will be awesome thank you yeah now i really appreciate it uh thanks to to both of you guys for for inviting us uh thanks of course to everybody who's uh watching this or reading transcripts or you know connected to them to the mind meld and you know getting an instant download of all humanity's knowledge 200 years from now uh and yeah i mean as i said i think the ultimate purpose of all of this is to be useful and it's an evolving conversation so we're really curious to to hear thoughts questions uh other ideas that might pop up and yeah i mean i don't know if your emails are available but if not you can make them available you can reach out to us on twitter or down to to the active inference labs organizers and they can connect us yeah same here thank you both of you and thank you to serval and dean for the questions uh we are i think both of us are available on

twitter or otherwise to have questions would love to follow up on certain things you know and see if we can make interesting projects out of it thank you folks this has been fun thank you
bye cheers bye good times